

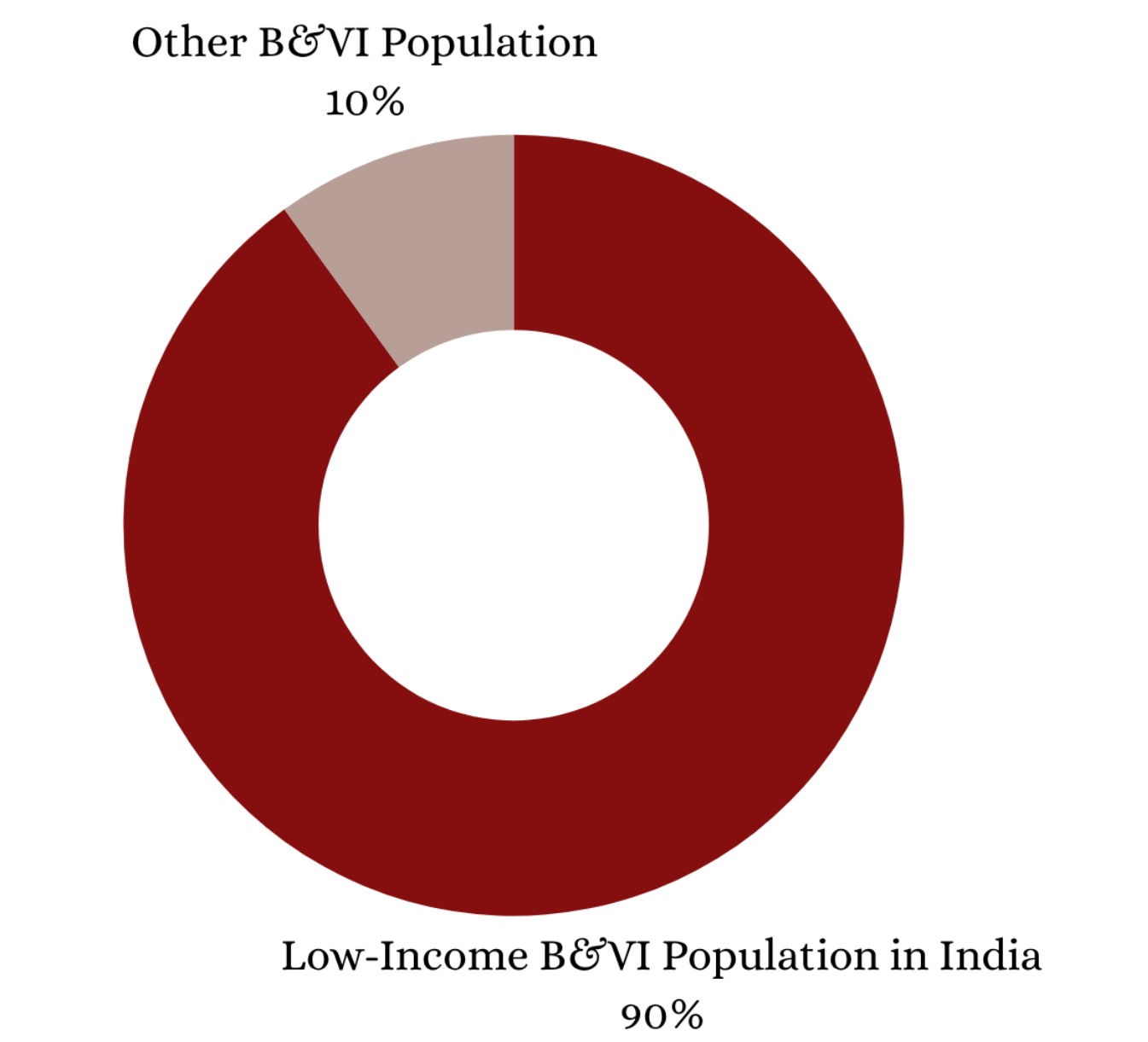
Dots: A Frugal Real-Time Tactile Display for Accessible STEM Education

Vaanee Tripathi ¹

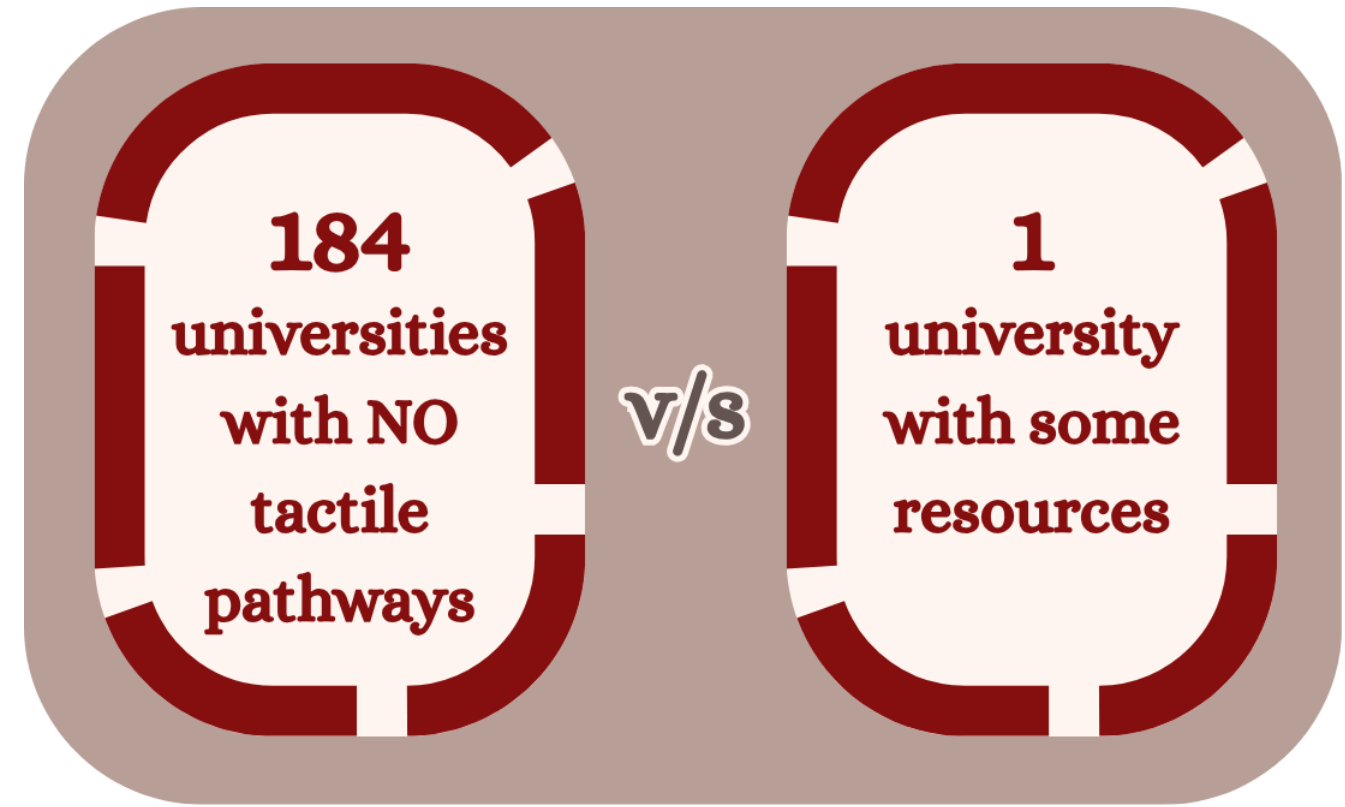
¹Ashoka University

Background

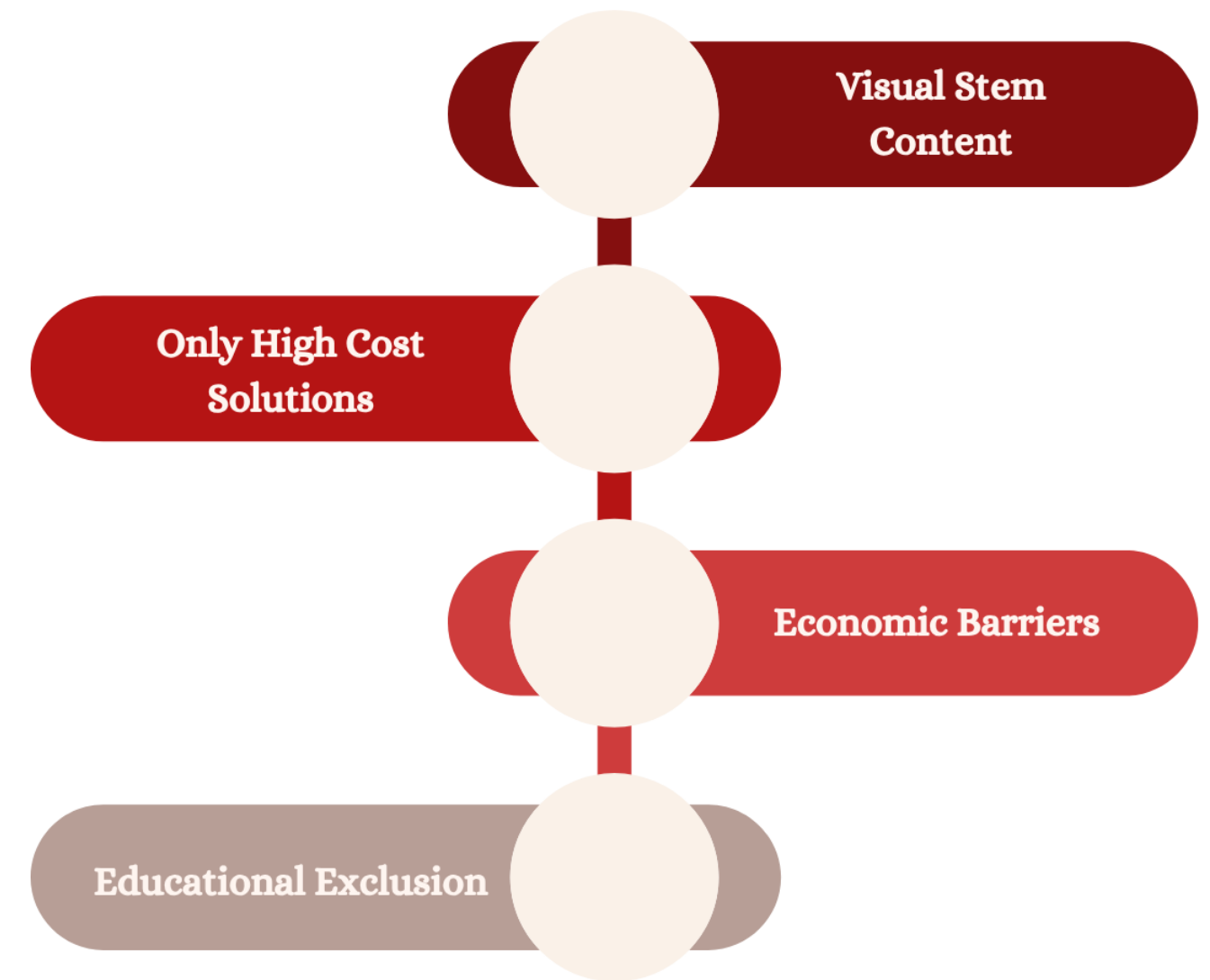
STEM education relies heavily on visual representations—graphs, diagrams, and spatial models—making it inherently inaccessible to students with visual impairments. Commercial tactile displays cost \$12,000-\$22,500, creating insurmountable economic barriers for educational institutions. With 90% of India’s visually impaired population coming from low-income families, these high costs effectively exclude an entire demographic from STEM education and career opportunities



As a result, only 1 of 185 surveyed Indian universities (0.54%) provides tactile STEM pathways.



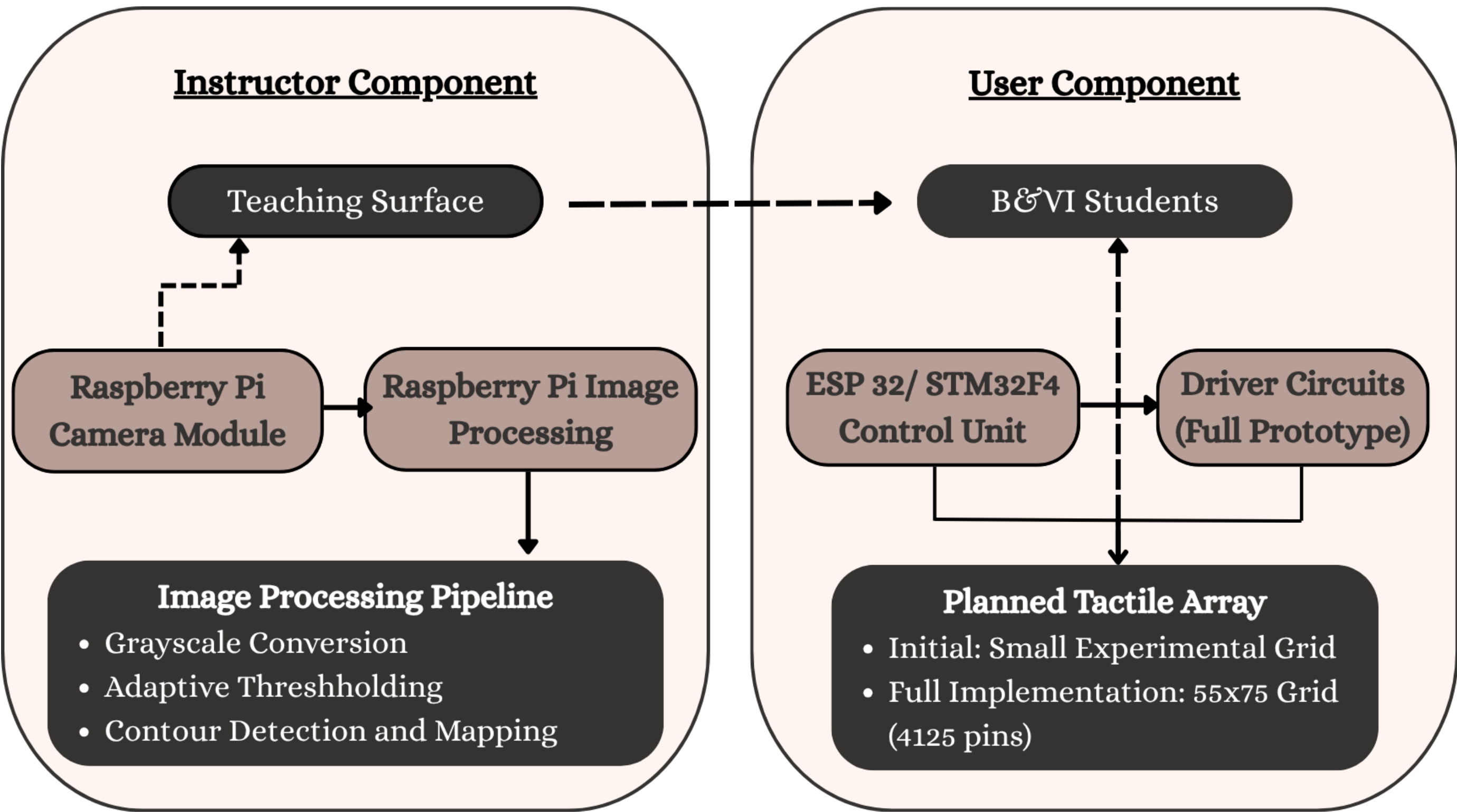
This systematic exclusion denies visually impaired students access to STEM careers and the economic opportunities they provide, perpetuating cycles of inequality and limiting India's potential to leverage its full talent pool in critical STEM fields.



Design

The project here discusses and proposes the following solution.

A frugal open source refreshable tactile dot grid with synchronous learning. The system architecture employs a dual-component approach designed for real-time tactile feedback. The instructor-focused component captures and processes visual information from teaching surfaces, while the student-focused component provides tactile output.

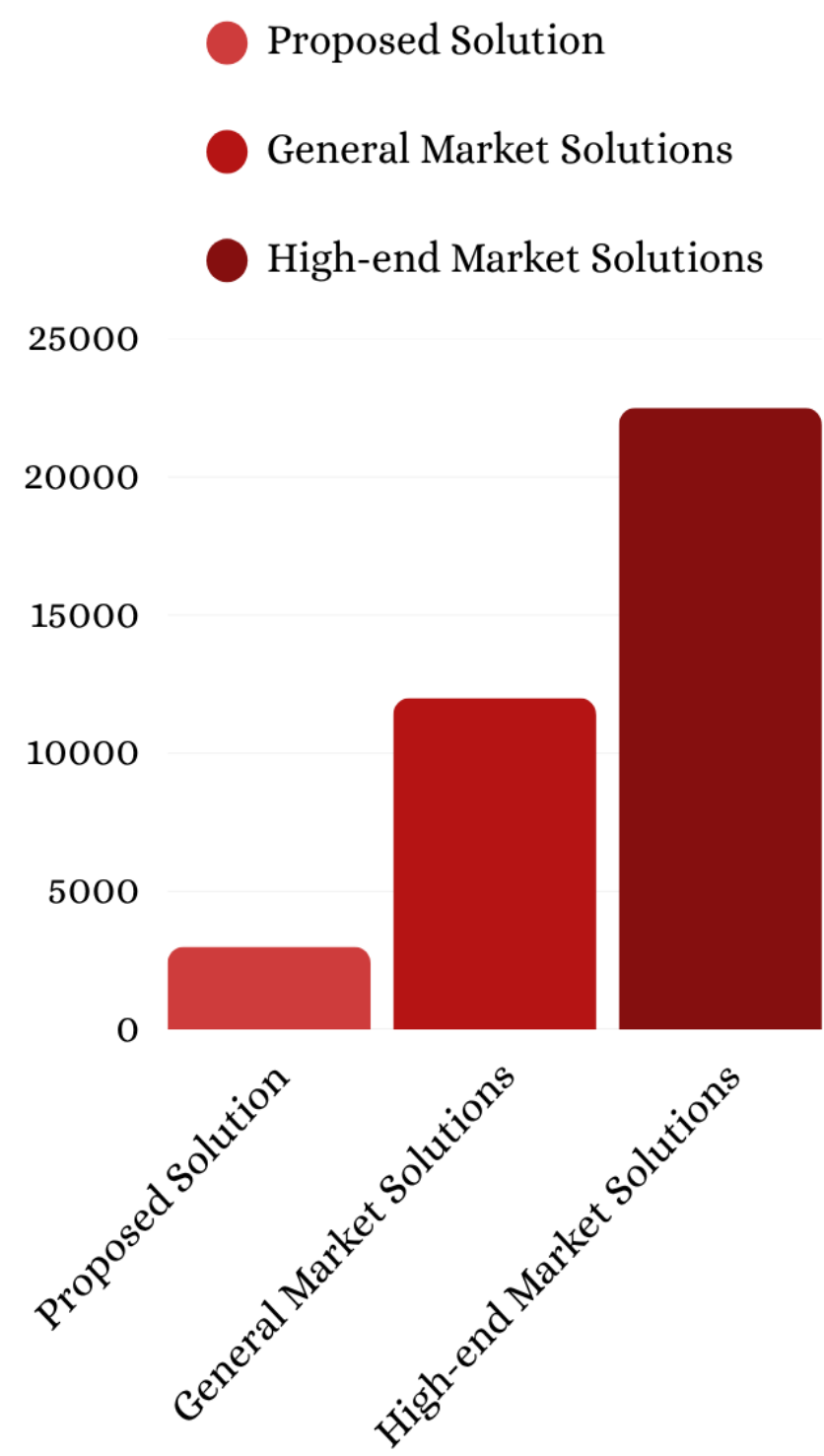


The system employs a comprehensive two-part approach to transform visual STEM content into tactile feedback. The first stage implements a real-time processing pipeline that captures visual content from teaching surfaces using a camera, enhances image quality through filtering and contrast adjustment, identifies key visual elements through edge detection, maps these elements to a grid format matching the tactile display dimensions, and wirelessly transmits the resulting data.

The second stage features the planned hardware architecture for the full 55×75 tactile array, utilizing a coordinated control system where a central microcontroller works with specialized circuits to manage 4,125 individual tactile elements, enabling each dot to be independently raised or lowered through bidirectional control mechanisms, supported by a power management system designed to handle the electrical demands of simultaneous actuator operation.

Discussion

This design addresses this accessibility crisis with a comprehensive tactile display system costing approximately \$3,000—achieving a 75% cost reduction compared to commercial alternatives. This dramatic price decrease ensures that STEM education is no longer held hostage by hefty bills that exclude entire populations from accessing critical learning opportunities and career pathways.



Several key questions remain to be answered about the system’s real-world impact and feasibility. Does synchronous tactile learning actually improve student outcomes compared to traditional methods? Can the cost be reduced even further to reach more institutions? How can the design be made more robust to work effectively in typical classroom conditions without requiring perfect lighting or precise positioning? The next phase involves building a complete prototype and conducting thorough user testing with visually impaired students to validate both the technology’s performance and its educational effectiveness in practice.

References

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